

Q. Show that series

$$\frac{1}{x} + \frac{2!}{x(x+1)} + \frac{3!}{x(x+1)(x+2)} + \dots$$

converges if $x > 2$ and
if $x < 2$.

Sol:-

$$U_n = \frac{n!}{x(x+1)(x+2)\dots(x+n-1)}$$

$$U_{n+1} = \frac{(n+1)!}{x(x+1)(x+2)\dots(x+n-1)(x+n)}$$

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$$\left| \frac{U_{n+1}}{U_n} \right| = \frac{(n+1)!}{x(x+1)(x+2)\dots(x+n-1)(x+n)} \cdot \frac{x(x+1)(x+2)\dots(x+n-1)}{n!}$$

$$= \left| \frac{(n+1)}{(x+n)} \right|$$

$$\lim_{n \rightarrow \infty} \frac{n \left(1 + \frac{1}{n} \right)}{x \left(\frac{x}{n} + 1 \right)} = 1 \text{ Test inconclusive.}$$

Now, Helping Test

$$n \left(\frac{U_n}{U_{n+1}} - 1 \right)$$

$$n \left(\frac{\cancel{n!} \cdot x \cdot x(x+1)(x+2) \dots (x+n-1)(x+n) - 1}{x(x+1)(x+2) \dots (x+n-1) \cdot \underbrace{(n+1)!}_{\cancel{n!}(n+1)}} - 1 \right)$$

$$n \left(\frac{x+n}{n+1} - 1 \right)$$

$$n \left(\frac{x+n - (n+1)}{n+1} \right) = n \left(\frac{x + \cancel{n} - \cancel{n} - 1}{n+1} \right) = n \left(\frac{x-1}{n+1} \right) = \cancel{n} \left(\frac{x-1}{\cancel{n}(1+\frac{1}{n})} \right)$$

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Now limit
 $n \rightarrow \infty$

$$\frac{x-1}{1+0} = x-1$$

$L = x-1$
for converge
 $L > 1$

$$x-1 > 1$$
$$x > 2 \text{ prove}$$

for diverge
 $L < 1$

$$x-1 < 1$$
$$x < 2 \text{ prove}$$

$x-1$